

Shor's algorithm TEMPORAL NAME

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An article usually includes an abstract, a concise summary of the work covered at length in the main body of the article. It is used for secondary publications and for information retrieval purposes. For AQM, the abstract should be at most 8 lines and the text (excluding authors, abstract but including tables, figures and references) should be less than 4 pages (about 3500 words). A set of standard references can be found in the `biblio.bib` file, which is compiled using BibTeX.

Sergio: I didn't know if you wanted to keep the whole template, I leave after this section

sical proposal.

INTRODUCTION

DAVID: Quantum advantage refers to the ability of quantum devices to outperform classical systems in solving certain problems. Although the term itself has gained popularity only recently, Richard Feynman already suggested in 1986 that exploiting quantum mechanics could enable more powerful computation than its classical counterpart [9]. In the early 1990s, several scientists, including Deutsch, Jozsa, Berthiaume, and Brassard, explicitly addressed this idea, showing that while certain tasks could be solved exactly using quantum mechanics, their classical analogues required randomization and could only be solved with high probability [5–7]. However, these works did not provide a problem outside the class of those already known to be solvable in polynomial time on a classical computer with the aid of randomization and small error [16]. In 1994, inspired by Bernstein and Vazirani's work [4], Daniel Simon made a groundbreaking contribution: he identified an oracle problem solvable in polynomial time on a quantum computer but requiring exponential time classically [15], thereby providing a concrete confirmation of Feynman's intuition.

The possibility of quantum computing excelling beyond classical one is a looming threat to, among other fields, cryptography, as some of its methods are often relied on hard-to-solve computational problems. This is the case for the widely used RSA public-key scheme developed by Rivest, Shamir and Adleman in 1978 [12]. In this system, the security rests on the difficulty of factorizing an integer into prime numbers. In 1995, the fastest known algorithm for factorizing an integer was Number Field Sieve by Lenstra, Lenstra Jr., Manasse and Pollard [11]. Such method takes an asymptotic running time $O(\exp(c(\log n)^{\frac{1}{3}}(\log \log n)^{\frac{2}{3}}))$ to factor an integer n and for some constant c .

In this article, Shor introduces the first quantum algorithm for integer factorization with an asymptotic running time of $O((\log n)^2(\log \log n)(\log \log \log n))$, demonstrating a clear quantum advantage over Lenstra's clas-

MATHEMATICAL BACKGROUND OF THE ALGORITHM

In this section, we address how to reduce the factorization problem to a modular exponentiation problem.

SERGIO: The goal of the algorithm is the following: given a positive integer n , find $m, t \in \mathbb{N}$, where $m, t > 1$, such that $n = mt$, that is, to decompose n as a product of two non-trivial positive integers. It has been later showed that n can be assumed to be neither odd or a prime power, as for the first case 2 is a trivial factor and an efficient classical algorithm can be found for the latter[1, 3].

Let $\tilde{m}$ (I rather use another variable, same as Wikipedia a or same as article x) be a candidate divisor of n , chosen such that $1 < \tilde{m} < n$. Using the Euclidean algorithm, one can compute $\gcd(\tilde{m}, n)$ in polynomial time [14].

If $\gcd(\tilde{m}, n) \neq 1$, then we have found a non-trivial divisor of n . In this case, set $m = \gcd(\tilde{m}, n)$ and $t = \frac{n}{\gcd(\tilde{m}, n)}$.

Otherwise, if $\gcd(\tilde{m}, n) = 1$, then the equivalence class $[\tilde{m}]$ belongs to the multiplicative group,

$$\left(\frac{\mathbb{Z}}{n\mathbb{Z}}\right)^* = \{[a] \in \frac{\mathbb{Z}}{n\mathbb{Z}} : \gcd(a, n) = 1\}.$$

In this case, we look for the *order* of $[\tilde{m}]$, namely the smallest positive integer r such that $\tilde{m}^r \equiv 1 \pmod{n}$.

This order r always exists since $\left(\frac{\mathbb{Z}}{n\mathbb{Z}}\right)^*$ is a finite abelian group. The efficient computation of r is precisely the step where the quantum part of the algorithm becomes essential.

Once r is obtained, two cases arise. If r is odd, the choice of $\tilde{m}$ is not useful, and the procedure is repeated with a new guess $\tilde{m}$. If r is even, then

$$(\tilde{m}^{r/2} - 1)(\tilde{m}^{r/2} + 1) \equiv 0 \pmod{n}.$$

Since r is minimal, we know that $\tilde{m}^{r/2} \not\equiv 1 \pmod{n}$. Thus it follows that $\tilde{m}^{r/2} \equiv -1 \pmod{n}$ i.e. $\tilde{m}^{r/2} + 1$ is a multiple of n . Therefore, we can compute $\gcd(\tilde{m}^{r/2} + 1, n)$. If $\gcd(\tilde{m}^{r/2} + 1, n) \neq 1$, then $\gcd(\tilde{m}^{r/2} + 1, n)$ is a non-trivial divisor of n , and the factorization goal is

achieved. Otherwise, the procedure is restarted with a new choice of $\tilde{m}$.

Sergio: Here there is the template

INTRODUCTION

This sample document demonstrates proper use of REVTeX 4 (and L^AT_EX 2_ε) in manuscripts prepared for submission. Further information can be found in the REVTeX 4 documentation included in the distribution or available at <http://publish.aps.org/revtex4/>.

When commands are referred to in this example file, they are always shown with their required arguments, using normal T_EX format. In this format, #1, #2, etc. stand for required author-supplied arguments to commands. For example, in `\section{#1}` the #1 stands for the title text of the author's section heading, and in `\title{#1}` the #1 stands for the title text of the paper.

Line breaks in section headings at all levels can be introduced using `\\`. A blank input line tells T_EX that the paragraph has ended. Note that top-level section headings are automatically uppercased. If a specific letter or word should appear in lowercase instead, you must escape it using `\lowercase{#1}` as in the word “via” above.

SECOND-LEVEL HEADING: FORMATTING

The `widetext` environment will make the text the width of the full page. The width-changing commands only take effect in `twocolumn` formatting. It has no effect if `preprint` formatting is chosen instead.

To cite bibliography entries, use the `\cite{#1}` command. Most journal styles will display the corresponding number(s) in square brackets: [10]. To avoid the square brackets, use `\onlinecite{#1}`: Refs. 10 and 8, 13. REVTeX “collapses” lists of consecutive reference numbers where possible. We now cite everyone together [2, 8, 13], and once again (Refs. 2, 8, 13). Note that the references were also sorted into the correct numerical order as well.

Footnotes are produced using the `\footnote{#1}` command. Some journal styles put footnotes into the bibliography. REVTeX 4 does this as well, but instead of interleaving the footnotes with the references, they are listed at the end of the references. Because the correct numbering of the footnotes must occur after the numbering of the references, an extra pass of L^AT_EX is required in order to get the numbering correct.

FORMATTING

Equations

Inline math may be typeset using the `$` delimiters. Bold math symbols may be achieved using the `\bm` package and the `\bm{#1}` command it supplies. For instance, a bold α can be typeset as `\bm{\alpha}` giving α . Fraktur and Blackboard (or open face or double struck) characters should be typeset using the `\mathfrak{#1}` and `\mathbb{#1}` commands respectively. Both are supplied by the `amssymb` package. For example, `\mathbb{R}` gives $\mathbb{R}$ and `\mathfrak{G}` gives $\mathfrak{G}$.

In L^AT_EX there are many different ways to display equations, and a few preferred ways are noted below. Displayed math will center by default. Use the class option `fleqn` to flush equations left.

Below we have numbered single-line equations; which are the most common equations in this type of text:

$$\chi_+(p) \lesssim [2|\mathbf{p}|(|\mathbf{p}| + p_z)]^{-1/2} \begin{pmatrix} |\mathbf{p}| + p_z \\ px + ip_y \end{pmatrix}, \quad (1)$$

$$f(x) = x^2 + \psi_\alpha(x) \quad (2)$$

$$\left\{ 1234567890abc123\alpha\beta\gamma\delta1234556\alpha\beta \frac{1\sum_b^a}{A^2} \right\}. \quad (3)$$

Note the open one in Eq. (3).

Not all numbered equations will fit within a narrow column this way. The equation number will move down automatically if it cannot fit on the same line with a one-line equation:

$$\left\{ ab12345678abc123456abcdef\alpha\beta\gamma\delta1234556\alpha\beta \frac{1\sum_b^a}{A^2} \right\}. \quad (4)$$

When the `\label{#1}` command is used [cf. input for Eq. (3)], the equation can be referred to in text without knowing the equation number that T_EX will assign to it. Just use `\ref{#1}`, where #1 is the same name that used in the `\label{#1}` command.

Unnumbered single-line equations can be typeset using the `\[, \]` format:

$$g^+g^+ \rightarrow g^+g^+g^+g^+ \dots, \quad q^+q^+ \rightarrow q^+g^+g^+ \dots$$

Figures

Figures may be inserted by using either the `graphics` or `graphicx` packages. These packages both define the `\includegraphics{#1}` command, but they differ in how optional arguments for specifying the orientation, scaling, and translation of the figure. Fig. 1 shows a figure that is small enough to fit in a single column. It is embedded using the `figure` environment which provides both the caption and the imports the figure file.

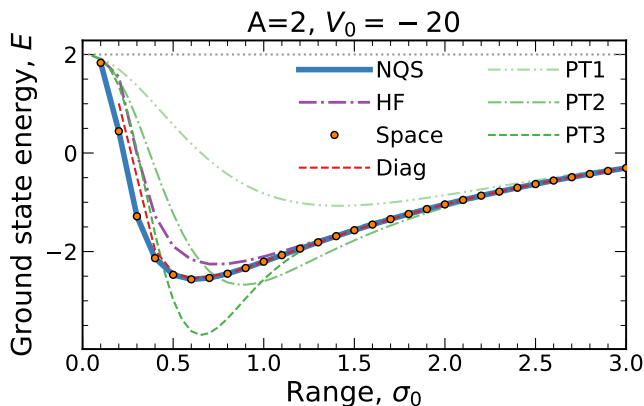


FIG. 1: A figure caption. The figure captions are automatically numbered. When preparing figures, use vector formats (pdf, eps), make sure labels are readable and that lines are differentiated not only by color, but also employing different linestyles.

Fig. 2 is a figure that is too wide for a single column, so instead the `figure*` environment has been used.

Tables

The heart of any table is the `tabular` environment which gives the rows of the tables. Each row consists of column entries separated by `&`'s and terminates with `\\`. The required argument for the `tabular` environment specifies how data are displayed in the columns. For instance, entries may be centered, left-justified, right-justified, aligned on a decimal point. Extra column-spacing may be specified as well, although REVTeX 4 sets this spacing so that the columns fill the width of the table. Horizontal rules are typeset using the `\hline` command. The doubled (or Scotch) rules that appear at the top and bottom of a table can be achieved enclosing the `tabular` environment within a `ruledtabular` environment. Rows whose columns span multiple columns can be typeset using the `\multicolumn{#1}{#2}{#3}` command (for example, see the first row of Table III).

Tables I-IV show various effects. Tables that fit in a narrow column are contained in a `table` environment. Table III is a wide table set with the `table*` environment. Long tables may need to break across pages. The most straightforward way to accomplish this is to specify the `[H]` float placement on the `table` or `table*` environment. However, the standard L^AT_EX 2_ε package `longtable` will give more control over how tables break and will allow headers and footers to be specified for each page of the table. A simple example of the use of `longtable` can be found in the file `summary.tex` that is included with the REVTeX 4 distribution.

There are two methods for setting footnotes within a

TABLE I: This is a narrow table which fits into a narrow column when using `twocolumn` formatting. Note that REVTeX 4 adjusts the intercolumn spacing so that the table fills the entire width of the column. Table captions are numbered automatically. This table illustrates left-aligned, centered, and right-aligned columns.

Left ^a	Centered ^b	Right
1	2	3
10	20	30
100	200	300

^aNote a.

^bNote b.

table (these footnotes will be displayed directly below the table rather than at the bottom of the page or in the bibliography). The easiest and preferred method is just to use the `\footnote{#1}` command. This will automatically enumerate the footnotes with lowercase roman letters. However, it is sometimes necessary to have multiple entries in the table share the same footnote. In this case, there is no choice but to manually create the footnotes using `\footnotemark{#1}` and `\footnotetext{#1}{#2}`. `#1` is a numeric value. Each time the same value for `#1` is used, the same mark is produced in the table. The `\footnotetext{#1}{#2}` commands are placed after the `tabular` environment. Examine the L^AT_EX source and output for Tables I and II for examples.

Scientific writing style requires that the initial citation of figures or tables be in numerical order in text, so don't cite Fig. 2 until Fig. 1 has been cited. All figures should be cited in the main body of the text, otherwise you will create a floating figure that readers may not know how to interpret. The same holds for Tables, they should all

TABLE II: A table with more columns still fits properly in a column. Note that several entries share the same footnote. Inspect the L^AT_EX input for this table to see exactly how it is done.

	r_c (Å)	r_0 (Å)	κr_0		r_c (Å)	r_0 (Å)	κr_0
Cu	0.800	14.10	2.550	Sn ^a	0.680	1.870	3.700
Ag	0.990	15.90	2.710	Pb ^b	0.450	1.930	3.760
Au	1.150	15.90	2.710	Ca ^c	0.750	2.170	3.560
Mg	0.490	17.60	3.200	Sr ^d	0.900	2.370	3.720
Zn	0.300	15.20	2.970	Li ^b	0.380	1.730	2.830
Cd	0.530	17.10	3.160	Na ^e	0.760	2.110	3.120
Hg	0.550	17.80	3.220	K ^e	1.120	2.620	3.480
Al	0.230	15.80	3.240	Rb ^e	1.330	2.800	3.590
Ga	0.310	16.70	3.330	Cs ^d	1.420	3.030	3.740
In	0.460	18.40	3.500	Ba ^e	0.960	2.460	3.780
Tl	0.480	18.90	3.550				

^aHere's the first, from Ref. 10.

^bHere's the second.

^cHere's the third.

^dHere's the fourth.

^eAnd etc.

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